

Question 1: Select the best Answer, from the following (/12 Marks)

1.1. In a graphical system, an array of pixels in the picture are stored in which of the following locations?

- a. **Frame buffer**
- b. Processor
- c. Flash Memory
- d. All of the above

1.2. Geometry Shapes of a 3d object are made of a wireframe. The wireframe contains:

- a. Grid of Pixels that form images
- b. **Vertices, edges, and faces that form polygons**
- c. Frames to form animation
- d. None of the above

1.3. The transformation in which the size of an object can be modified in x-direction, y-direction, and z-direction

- a. Translation
- b. **Scaling**
- c. Rotation
- d. All of the above

1.4. The higher number of pixels gives us a ____ image

- a. **Better**
- b. Worst
- c. Smaller
- d. None of the above

1.5. Elements of rendering:

- a. Object
- b. Light
- c. Material
- d. Camera
- e. **All the above**

1.6. Diffuse lighting type is:

- a. A light that is scattered by the environment
- b. **A directional light cast made by a light source**
- c. A reflection of light on surfaces
- d. None of the above

1.7. What is visual texture?

- a. It is the painting of glass objects made to look transparent and shiny
- b. The creation of realistic-looking textures in a work
- c. The use of delicate brush strokes or other marks to create fine lines

d. All of the above

1.8. Lighting model involves a number of factors

- a. Light attributes
- b. Object surface attributes
- c. Object position and object orientation
- d. All of the above**

1.9. In which 3D projection do objects that are far away from the camera appear smaller compared to those nearby?

- a. Perspective projection**
- b. Orthographic projection
- c. Projection that has no convergence point
- d. None of the above

1.10. Parallel/Orthographic projection shows the

- a. True image of an object
- b. True size of an object
- c. True shape of an object
- d. All of the above**

1.11. What happens if the values of scaling factors s_x and s_y are less than 1 (i.e., $s_x < 1$ and $s_y < 1$)?

- a. No changes in the object's size
- b. Increase the object's size
- c. Reduce the object's size**
- d. None of the above

1.12. Pixel can be arranged in a regular

- a. One-dimensional grid
- b. Two-dimensional grid**
- c. Three-dimensional grid
- d. None of the above

1.13. Each pixel has _____ basic color components

- a. Two or three
- b. One or two
- c. Three or four**
- d. None of the above

1.14. RGB true color model has _____ color depth

- a. 24-bit**
- b. 32-bit
- c. 64-bit
- d. None of the above

1.15. Ambient lighting type is:

- a. A light that is scattered by the environment**
- b. A directional light cast made by a light source

- c. A reflection of light on surfaces
- d. None of the above

1.16. The transparency for color is called the _____ and its value is a number between 0.0 and 1.0

- a. Alpha level
- b. Color gamut
- c. Mach bands
- d. Emissive color

1.17. How many vectors does a cube with six faces has?

- a. 3 Vectors
- b. 6 Vectors
- c. 8 Vectors
- d. 12 Vectors

1.18. A vertex in graphics modeling defines a:

- a. Point
- b. Line
- c. Polygon
- d. None of the above

1.19. What is the word that describes materials that light passes through easily?

- a. Transparent
- b. Opaque
- c. Translucent
- d. Redundant



1.20. When an object blocks the light, what is formed in the opposite direction of light?

- a. A reflection
- b. An illumination
- c. A shadow
- d. A refraction

1.21. When an object moves closer to a light source, its shadow ...

- a. Gets smaller
- b. Gets bigger
- c. Changes shape
- d. Stays the same size

1.22. In the attached figure, why does the person have three shadows?

- a. His position is in the modeling coordinates
- b. Environmental light is used
- c. Three light sources from different directions pointing to him
- d. Sun light source is used



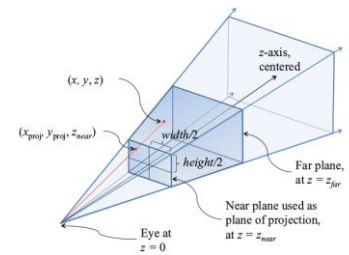
1.23. Which of the following OpenGL functions allow you to draw a polygon?

- a. GL_POINTS
- b. GL_LINE_LOOP

- c. GL_TRIANGLE
- d. All of the above

1.24. Which projection do the following object represent?

- a. Parallel projection
- b. Orthographic projection
- c. **Perspective projection**
- d. Scaled projection



Question 2: Write (T) for correct statements and F for the wrong ones

(/5 Marks)

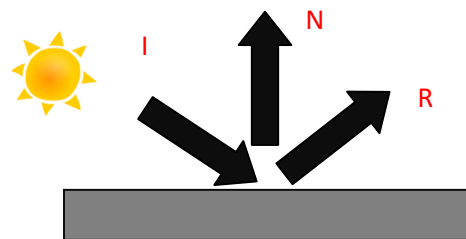
1. To simulate real-life lighting, it is not enough to only add a light source; other factors need to be considered as well **T**
2. The identity matrix is a square matrix with 0s on the diagonal and 1s elsewhere. **F**
3. In OpenGL, you can add a light source using the function glLight **T**
4. In OpenGL, the function glMaterial allows you to manage the properties of an object surface **F**
5. Texture mapping is a technique used to add detail to the surface of an object by mapping a 2D image onto a 2D or 3D object **T**
6. In OpenGL, the function glTexImage2D is used for animating 2D graphics **F**
7. An orthographic projection, projects all the objects in the scene to the viewing plane by parallel lines. **T**
8. In OpenGL, the clipping area is set by default -1 to +1 on each axis **T**
9. In OpenGL, the 4x4 matrix is used in two modes: model-view mode and projection mode **T**
10. In OpenGL, the viewport takes the same dimensions of the display window if it is not defined **T**

Question 3:

(?/5 Marks)

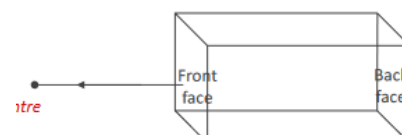
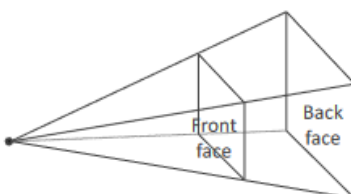
(a) Label each vector in the figure with an appropriate letter (I,N,R)

- I – the incident light ray
- N – the surface normal vector
- R – the reflected light ray



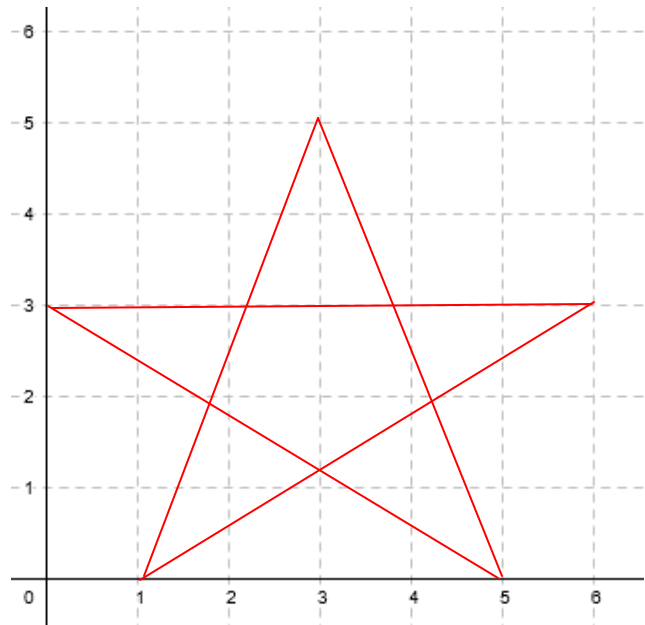
(b) Label the following two figures with:

- Parallel projection
- Perspective projection

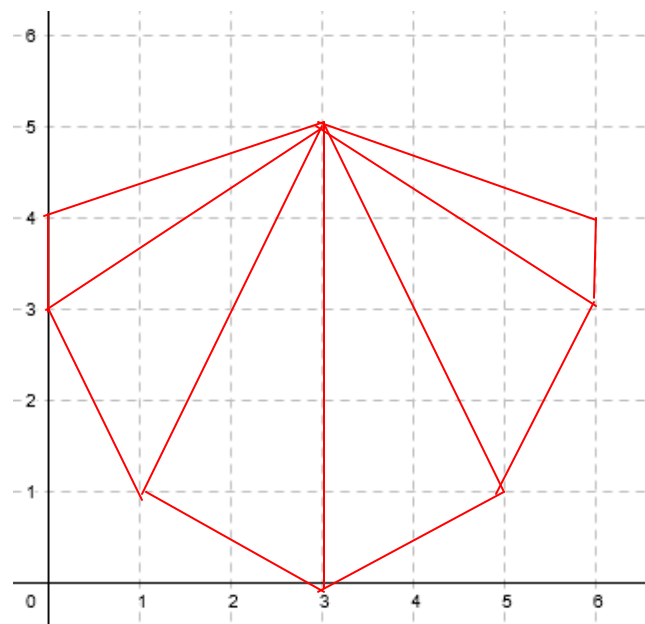


Question 4: Sketch the output of this OpenGL code by drawing any lines and shading any region (/4 Marks)

```
glBegin(GL_LINE_LOOP);
glVertex2i (1, 0);
glVertex2i (3, 5);
glVertex2i (5, 0);
glVertex2i (0, 3);
glVertex2i (6, 0);
glEnd();
```



```
glBegin(GL_TRIANGLE_FAN);
glVertex2i (3, 5);
glVertex2i (0, 4);
glVertex2i (0, 3);
glVertex2i (1, 1);
glVertex2i (3, 0);
glVertex2i (5, 1);
glVertex2i (6, 3);
glVertex2i (6, 4);
glEnd();
```



Question 5:**(/4 Marks)****Write the corresponding description of the following code:**

- Identity
- Translation
- Rotation (around axis)
- Scaling

(a) <code>return numpy.array([[1,0,0,0], [0,1,0,0], [0,0,1,0], [0,0,0,1]]).astype(float)</code>	Identity
(b) <code>c = cos(angle) s = sin(angle) return numpy.array([[1,0,0,0], [0,c,-s,0], [0,s,c,0], [0,0,0,1]]).astype(float)</code>	Rotation
(c) <code>return numpy.array([[2,0,0,0], [0,2,0,0], [0,0,1,0], [0,0,0,1]]).astype(float)</code>	Scaling
(d) <code>return numpy.array([[0,0,0,1], [0,0,0,1], [0,0,0,0], [0,0,0,1]]).astype(float)</code>	Translation

Question 6: Describe the function of the following codes**(/2 Marks)**

```
class Example(Base):
    def initialize(self):
        print("Initializing program...")
        self.renderer = Renderer()
        self.scene = Scene()
        self.camera = Camera(aspect_ratio=800/600)
        self.camera.set_position([0, 0, 1])
        geometry = RectangleGeometry()
        grid_texture = Texture(file_name="../images/main.jpg")
        material = TextureMaterial(grid_texture)
        self.mesh = Mesh(geometry, material)
        self.scene.add(self.mesh)
```

Textured Square.....

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```
class Example(Base):
    def initialize(self):
        print("Initializing program...")
        self.renderer = Renderer()
        self.scene = Scene()
        self.camera = Camera(aspect_ratio=800/600)
        self.camera.set_position([0, 0, 4])
        geometry = BoxGeometry()
        # material = PointMaterial(property_dict={"baseColor": [1, 1, 0], "pointSize": 5})
        material = SurfaceMaterial(property_dict={"useVertexColors": True})
        # material = SurfaceMaterial(
        #     property_dict= {
        #         "useVertexColors": True,
        #         "wireframe": True,
        #         "lineWidth": 8
        #     }
        # )
        self.mesh = Mesh(geometry, material)
        self.scene.add(self.mesh)
```

3D cube.....

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MAFIA FCIT20:

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